

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                      | Marks Available |          |          |           |          |          |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 4        | (a) | (i)  |  | Slow down neutrons (1)<br>So they {cause fission of uranium / can be captured by an uranium (atom) / can be captured by an uranium (nucleus)} (1)                                                                                                                                                                                                                                                    | 2               |          |          | 2         |          |          |
|          |     | (ii) |  | Any 3 × (1) from:<br>Control rods may no longer be free to move (up and down in the channels) (1)<br>If rods are stuck out then the reactor may start to overheat (1) since they will not be able to absorb {sufficient / enough} neutrons (to reduce rate of fission) (1)<br>leading to possible meltdown / uncontrollable chain reaction} / you can't shut the reactor down (1)<br><br>OR converse | 3               |          |          | 3         |          |          |
|          | (b) |      |  | $^{92}_{36}\text{Kr} + 3$ (1)                                                                                                                                                                                                                                                                                                                                                                        |                 | 3        |          | 3         | 2        |          |
|          | (c) | (i)  |  | Correct construction line(s) added (1)<br>700 million years (1)                                                                                                                                                                                                                                                                                                                                      |                 | 2        |          | 2         | 2        |          |
|          |     | (ii) |  | Sequence of halving – expect $1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{4} \rightarrow \frac{1}{8} \rightarrow \frac{1}{16} \rightarrow \frac{1}{32}$ (1)<br>5 half-lives (1)<br>$5 \times 700\,000\,000(\text{ecf}) = 3\,500\,000\,000$ years (1)                                                                                                                                              |                 | 3        |          | 3         | 3        |          |
|          |     |      |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                              | <b>5</b>        | <b>8</b> | <b>0</b> | <b>13</b> | <b>7</b> | <b>0</b> |